

AC / DC SOLID-STATE SWITCH + ISOLATED VOLTAGE SENSE

CIRCUIT DRAFT – VOLTAGE / CURRENT / LOAD TYPE REQUIRED BEFORE PCB DESIGN

1. LOGIC CONTROL – 3.3 V supply, active-high EN, both drivers always commanded together

2. BIDIRECTIONAL POWER PATH

Q1/Q2 sources share SOURCE_FLOAT.
Opposed body diodes block either polarity when OFF.
Gates are separate; driver outputs are not paralleled.
SOURCE_FLOAT must NOT connect to LOAD_RETURN.

No gate-source bleed resistor: the Si8751 has a high source impedance. A conventional 10k/100k pull-down would collapse its generated gate voltage.
Miller clamp connects through the drain capacitor.

Switching rate, surge suppression and thermal design remain pending the load specification.
No fabrication or mains rating assigned to this draft.

3. ISOLATED MEASUREMENT POWER – converter return is LOAD_RETURN

4. SWITCHED OUTPUT MEASUREMENT

OUT = VREF/2 at zero input AND when isolated supply is missing. Zero output indication is not proof of a de-energized rail.

Unrated circuit draft – power specification required

Sheet: /
File: Si8751AB-IS_devboard.kicad_sch

Title: Isolated AC/DC switch development board

Size: A3
KiCad E.D.A. 10.0.0

Date: 2026-09-04

Rev: 0.1-DRAFT
Id: 1/1